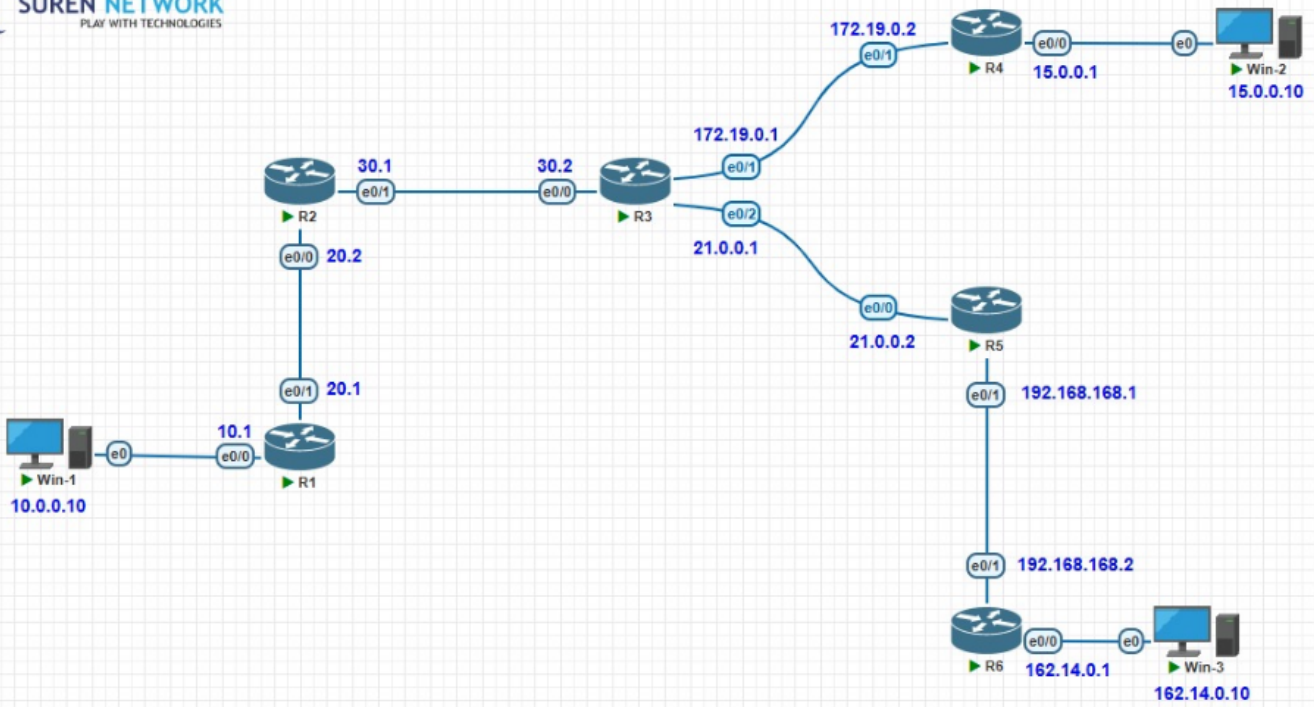




 R1

```
%Inconsistent address and mask
R1(config)#ip address 172.19.0.0 255.0.0.0 20.0.0.2
      ^
% Invalid input detected at '^' marker.

R1(config)#ip address 172.19.0.0 255.255.0.0 20.0.0.2
      ^
% Invalid input detected at '^' marker.

R1(config)#ip route 172.19.0.0 255.255.0.0 20.0.0.2
R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 21.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 192.168.168.0 255.255.255.0 20.0.0.2
R1(config)#ip route 162.14.0.0 255.255.0.0 20.0.0.2
R1(config)#exit
R1#
*Jul 20 08:41:09.944: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
```

R2

Ethernet0/3 unassigned YES TFTP down down

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 172.19.0.0 255.255.0.0
*Jul 20 08:54:25.030: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 21.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#exit
R2#
*Jul 20 08:58:19.796: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.19.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
```

R3

Ethernet0/2	21.0.0.1	YES manual up	up
Ethernet0/3	unassigned	YES TFTP down	down

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 15.0.0.0 255.0.0.0 172.19.0.2
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.2
R3(config)#exit
R3#
*Jul 20 09:13:30.582: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```



Ethernet0/3 unassigned YES TFTP down down

```
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route
*Jul 20 09:20:41.383: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#exit
R4#
*Jul 20 09:26:03.792: %SYS-5-CONFIG_I: Configured from console by console
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

R5

Ethernet0/1	192.168.168.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1

R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2

R5(config)#exit

R5#s

*Jul 20 09:47:32.280: %SYS-5-CONFIG_I: Configured from console by console

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 21.0.0.1

S 15.0.0.0/8 [1/0] via 21.0.0.1

S 20.0.0.0/8 [1/0] via 21.0.0.1

21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

21.0.0.0/8 is directly connected, Ethernet0/0

L 21.0.0.2/32 is directly connected, Ethernet0/0

S 30.0.0.0/8 [1/0] via 21.0.0.1

S 162.14.0.0/16 [1/0] via 192.168.168.2

S 172.19.0.0/16 [1/0] via 21.0.0.1

192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.168.0/24 is directly connected, Ethernet0/1

L 192.168.168.1/32 is directly connected, Ethernet0/1

R5#

R6

```
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#ip route 10.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 10.0.0.0 255.0.0.0 192.168.168.1
*Jul 20 09:53:49.018: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R6(config)#ip route 20.0.0.0 255.0.0.0 192.168.168.1
^
% Invalid input detected at '^' marker.

R6(config)#ip route 20.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 30.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 21.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 172.19.0.0 255.255.0.0 192.168.168.1
R6(config)#ip route 15.0.0.0 255.0.0.0 192.168.168.1
R6(config)#exit
R6#sho
*Jul 20 09:56:34.272: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
     162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
       C    162.14.0.0/16 is directly connected, Ethernet0/0
       L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
       C    192.168.168.0/24 is directly connected, Ethernet0/1
       L    192.168.168.2/32 is directly connected, Ethernet0/1
```

QEMU (Win-1)



C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	
IP Address	:	10.0.0.10
Subnet Mask	:	255.0.0.0
Default Gateway	:	10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=4ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=7ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>

QEMU (Win-2)



C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address . . . . . : 15.0.0.10  
Subnet Mask . . . . . : 255.0.0.0  
Default Gateway . . . . . : 15.0.0.1
```

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

```
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
```

Ping statistics for 10.0.0.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

```
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
```

Ping statistics for 162.14.0.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

C:\Documents and Settings\user>_

```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 162.14.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 162.14.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```